

ROGER D. LAUNIUS

APOLLO'S LEGACY

PERSPECTIVES ON
THE MOON LANDINGS



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Versions of Reality

Pete Conrad, the sole Apollo astronaut who had received an Ivy League education, spent the rest of his life trying to live it down. Even his entry onto the lunar surface, as part of the Apollo 12 mission on November 19, 1969, was marked by a sort of insistent folksiness. After making a precision landing within 600 feet of a robotic soft lander that had reached the Moon in 1967, he exclaimed to the listening world, in sheer excitement, "Whoopie! Man, that may have been a small one for Neil, but that's a long one for me." Thus he entered the public's perception in a way strikingly different from the way that the astronauts of the first Moon landing had been perceived. While some decried his exclamations as unbecoming of a national representative who was speaking to, and for, the entire world, others enjoyed his words and even saw him as a sort of folk hero. That perspective was reinforced when Conrad, in later recollections, said he was just happy that he hadn't cursed. Brief as this episode was, it suggests the divergent ways in which Apollo has been presented to the public and remembered in the years since humans last set foot on the Moon.

Another story—this one an old baseball joke—can also help us to understand the various ways we have talked about Apollo. Three umpires are discussing how they call balls and strikes behind the plate. One umpire says, "I call them as they are." The second says, "I call them as I see them." The third says, "They ain't nothin' 'til I call them."

The first umpire's view of the nature of reality, what it is, and who decides reflects a premodern, absolutist position: reality exists, he says, beyond the ways that any bystanders might perceive or interpret it. The second umpire's position, one of rationality and modernity, suggests that reality is shaped by perceptions. The third umpire's statement expresses a postmodern, existential belief about reality: what actually happened does not matter all that much, and the thing that is truly important is the decision he makes about that reality's meaning. It seems that this last perspective has become the critical element in the ways Americans consider

the Moon landings. What we think of Apollo is an intensely personal matter, one predicated on our many idiosyncrasies and perspectives.

Without question, Apollo was for most observers, and perhaps consistently for much of the American public, an epochal event that signaled the opening of a new frontier where a grand visionary future for Americans might be realized. It epitomized, most Americans have consistently believed, what sets the United States apart from the rest of the nations of the world. American exceptionalism reigns in this telling of Apollo, and the program is often depicted as a momentous undertaking in US history, one that must be revered because it shows how successful Americans can be when they try.

Apollo has been consistently viewed as a feel-good triumph for America and its people, an iconic moment of national accomplishment. An astronaut on the Moon saluting the American flag served well as a patriotic symbol, one made all the more powerful because it required Americans to overcome an existential threat from the Soviet Union. Celebrants of Apollo, accordingly, have long argued that returns on investment in the age of space exploration changed Americans' lives for the better. As President Lyndon B. Johnson remarked in response to successes in the Moon race in August 1965, "Somehow the problems which yesterday seemed large and ominous and insoluble today appear much less foreboding." Why should Americans fear problems on Earth, he believed, when they had accomplished so much in space? If we can go to the Moon, why can't we solve our other problems? In this story, the Moon landings demonstrate that we can accomplish anything we set our minds to. After July 20, 1969, "If we can put a man on the Moon, why can't we do X?" became a catchphrase, with the last part of the sentence filled in by any number of ambitious goals. Apollo fits beautifully into a long-standing desire to embrace American awesomeness.

This narrative found expression not just in the United States, but also around the world. One of the objectives of the Apollo program, as noted earlier, was to demonstrate American technological superiority as a means to bring allies into the Western camp during the Cold War, and it achieved this goal. Official congratulations poured in to the US president from other heads of state after the first landing, even as informal ones went to NASA and the astronauts. Every nation that had regular diplomatic relations with the United States sent its best wishes in recognition

of the success of the mission. By providing evidence of American technological prowess via the Moon landings, NASA served a valuable foreign policy objective, for it helped keep other nations in America's corner during the Cold War.

A counter-story to this dominant interpretation of American triumph and exceptionality also emerged in the 1960s: some argued that Apollo deserved criticism as a "moondoggle," a wasteful expenditure of federal funds that could have been spent much more effectively elsewhere. For example, Vannevar Bush, a well-respected leading scientist who appreciated the marshaling of the federal government's power in the furtherance of national objectives, wrote to NASA Administrator James E. Webb in April 1963 to voice his concerns about the cost versus the benefits of the Moon program. He asserted that Apollo, "as it has been built up, is not sound," and he expressed concern that it would prove "more expensive than the country can now afford," adding that "its results, while interesting, are secondary to our national welfare."

Sociologist Amitai Etzioni was even more critical. In a reasoned, full-length critique of Apollo in 1964, he deplored the "huge pile of resources" spent on space, "not only in dollars and cents, but the best scientific minds—the best engineering minds were dedicated to the space project." Could not those resources have been better spent on improving the lives of people in modern America? Etzioni bemoaned the nation's penchant for embracing both high technology and unsustainable materialism: "We seek to uphold humanist concerns and a quest for a nobler life under the mounting swell of commercial, mechanical, and mass-media pressure."

Several political leaders in the United States, especially in the Democratic Party, found that support for Apollo clashed with their support of funds for social programs enacted through Great Society legislation. They disparaged Apollo as being both too closely linked to the military-industrial complex and defense spending and too far removed from the ideals of racial, social, and economic justice at the heart of the positive liberal state that most Democrats envisioned. Senators such as J. William Fulbright, Walter Mondale, and William Proxmire challenged the Johnson administration every year over funding for Apollo that they believed could be more effectively used for social programs. Bureau of the Budget Director Charles Schultze worked throughout the mid-1960s to shift funds away from Apollo toward programs such as the war on

poverty. Meanwhile Johnson tried to defend Apollo as a part of his Great Society initiatives, arguing that it helped poor Southern communities through its infusion of federal investment into high-technology development programs located there. This proved to be a difficult sell, and the NASA budget declined precipitously throughout the latter half of the 1960s.

This perspective views Apollo as a waste, a missed opportunity to further important and necessary goals in America. Still, the triumphalist narrative of Apollo has been so powerful a memory that today most people in the United States believe that the program enjoyed universally enthusiastic support during the 1960s and that somehow NASA lost its compass thereafter. In fact, at only one point before the Apollo 11 mission, in October 1965, did more than half of the public favor the program. In most public opinion polls of the 1960s, Americans consistently ranked spaceflight near the top of those programs they wished to see cut in the federal budget; a majority ranked the space program as the government initiative most deserving of reduction and said its funding should be redistributed to Social Security, Medicare, and other social programs. While most Americans did not oppose Apollo per se, they certainly questioned spending on it during a time when social problems appeared more pressing.

At the same time, some figures on the American political right during the Apollo era criticized the program as an abuse of federal power. In their view, the federal government should not do much of anything, let alone put humans on the Moon; these critics held a persistently libertarian position that emphasized individual prerogative and freedom over state action. One such critic, former President Dwight D. Eisenhower, believed that empowering NASA to accomplish the Apollo Moon landings was a mistake, remarking in a 1962 article, "Why the great hurry to get to the moon and the planets? We have already demonstrated that in everything except the power of our booster rockets we are leading the world in scientific space exploration. From here on, I think we should proceed in an orderly, scientific way, building one accomplishment on another." He later cautioned that the Moon race "has diverted a disproportionate share of our brain-power and research facilities from equally significant problems, including education and automation." Likewise, in the 1964 presidential election, the Republican candidate, Senator Barry

Goldwater, urged a reduction of the Apollo commitment to pay for national security initiatives.

After the Moon landings came off successfully, however, the American right largely retreated from high-profile criticism of Apollo, and it stayed quiet on the topic until the 1980s, when a full-scale assault on 1960s-era Democratic Great Society initiatives emerged as a central part of the conservative strain that emerged fully during the Reagan years. Conservatives reappraising the '60s—especially its social upheavals and the US defeat in Vietnam—castigated the decade as a time of failure in American politics. Thus a new conservative space policy, as well as a conservative space history, emerged in the 1980s to criticize Apollo.

Nothing expresses this right turn better than the rehabilitation of Eisenhower as president. In reinterpretations by some 1980s historians, including Pulitzer Prize-winning historian Walter A. McDougall (who critiqued Apollo as part of a larger assault on what he believed were “products of the maniacal 1960s”), Eisenhower emerges as the hero of the early space age, seeking to hold down expenditures, refusing to race the Soviet Union into space, and working to maintain traditional balances among policy, economics, and security. As historian of technology Alex Roland has pointed out, “McDougall pictures [Eisenhower] as standing alone against the post-Sputnik stampede, unwilling to hock the crown jewels in a race to the moon, confident that America’s security could be guaranteed without a raid on the Treasury, and concerned lest a space race with the Russians jeopardize America’s values and freedoms and drag us down to the level of the enemy.”

Such linkages of space policy to social policy may seem tenuous at first, but conservative Apollo critics disapproved of the power of the federal government and the state system to “intrude” into individual lives. McDougall—to further examine one exemplar of such critiques—also argued that the space age had given birth to a state of “perpetual technological revolution” due to the industry that arose to support Apollo’s incredibly complex set of machines. He focused upon the role of the government as a promoter of technological progress and argued that the United States, in its drive to respond to the Soviet challenge, had recreated the same type of command technocracy that the Soviets themselves had instituted. He saw this as a detriment to American society as a whole, concluding that the Moon race had led to a cooptation of individual

liberty in the pursuit of state power and saying that any assertion of state authority represented an equal degradation of individual freedom:

In these years the fundamental relationship between the government and new technology changed as never before in history. No longer did state and society react to new tools and methods, adjusting, regulating, or encouraging their spontaneous development. Rather, states took upon themselves the primary responsibility for generating new technology. This has meant that to the extent revolutionary technologies have profound second-order consequences in the domestic life of societies, by forcing new technologies, all governments have become revolutionary, whatever their reasons or ideological pretensions.

Once institutionalized, technocracy has not gone away, a fact that led McDougall to conclude that Apollo was enormously costly to the nation, and not just in public treasure. Like other critics of Apollo on the right, he bemoaned government activism and government power, something that conservatives tended to challenge even though most Americans of the era accepted at face value the benign nature of that power.

Finally, there is the third strand of the Apollo story. Almost from the point of the first Apollo missions, a small group of Americans has denied that the race to the Moon—and particularly the Moon landings—took place at all. This group seems to be expanding, and its narrative is rising in importance as the events of Apollo recede into history, aided by a younger generation with no personal memory of what happened in the space-race era and for whom distrust of government runs high. Only 18 percent of all Americans in December 2017 said they trusted the government in Washington to do what is right “always” or “most of the time,” and among those younger than 40, such trust was lower still. Jaded by many other government scandals, and by real conspiracies such as those that took the United States into war in Iraq and fostered an economic crisis in 2008, some members of society have found themselves susceptible to the myriad Moon hoax advocates.

Denying the Moon landings invokes a response of “crank” and “crackpot” from most listeners; indeed, these conspiracy theories deserve disdain, and most Americans, surveys consistently show, reject them. But

their adherents can be viewed as part of a postmodern interpretation of history: at some level, Americans today, like the umpire who says, "They ain't nothin' 'til I call them," do not see absolutes in our history. Instead, we see everything as constructed. Versions of the past have replaced earlier versions that once *seemed* to be true, even as questions are raised about who has the right—not to mention the power—to interpret the past. Many timeworn narratives about America's history have been dismantled or revised in recent decades, from the absence (or even celebration) of the genocide of Native Americans in stories of the country's founding to historian Frederick Jackson Turner's 1893 frontier thesis, which argued that the national character was forged in westward expansion and which held sway for many decades before it was critiqued and abandoned in the late 20th century. Meanwhile other conspiracy theories have become major elements of the memory of the nation—for example, the belief that John F. Kennedy was assassinated by some nefarious network involving key members of the national security establishment rather than by a single gunman. It is conceivable that such an erroneous reevaluation might one day reshape our view of the Moon landings, too.

In the context of Apollo's history, the duels among its four primary "stories"—the dominant celebratory account of the program; critiques from the left and from the right; and the conspiracy theories—have been, in essence, a battle for control of national memory. Was Apollo an exemplar of American uniqueness, a bright and shining moment in the nation's history to be remembered that way for all time? Or was it a waste of time, energy, and resources? Was it a mistake? Or was it all a hoax? To return to the story of the umpires: do we call it as it is, call it as we see it, or claim that it's nothing until we've called it in the first place?

CONCLUSION

Remembering Apollo

One of the hallmarks of recollections of the Apollo program among Americans has been proud nationalism and exceptionalism that often bordered on jingoism. That sense of nationalism is embarrassing in a postmodern, global age, and the elements of it that one still sees in recollections of Apollo suggests that Americans have yet to deal effectively with them. What were the ingredients of this unique form of nationalism, its origin as a Cold War response to Soviet space successes, and its role in promoting United States power to the rest of the world? Truly this nationalism involved an exclusionary and unique perspective on spaceflight and American technological virtuosity that found powerful expression in the nation's popular culture.

In a certain sense, our memory of Apollo is a myth, a "pattern of behavior" essentially held in common. This myth is not so much a fable or falsehood as it is a story, a kind of poetry, about events and situations that have great significance both for those involved and those that follow. Myths are, in fact, essential truths for the members of a cultural group who hold them, enact them, or perceive them. They are sometimes expressed in diffuse ideologies, but in literate societies such as the United States they are also embedded in historical narratives.

Memory, myth, and history are closely akin to each other; essentially they are stories that explain how things got to be the way they are. But common parlance suggests that memory is often faulty, myth is fiction, and only history is, or at least aspires to be, true. History to me, however, is an attempt to recount, model, or reconstruct the memory of the past for the purposes of the present. For a variety of reasons, such attempts are never completely successful. Thus, although few historians overtly do so, it is important to distinguish between history—the recounting of past events—and the past that is truly lost forever. History never fully or completely or accurately describes the past but instead attempts to develop approximate mental models or reconstructions of events. Different

cultures at different times formulated and presented their reconstructions of the past in strikingly different ways. Thus, it is highly dangerous to attempt to evaluate the relationship between another culture's concept of "history," our own concept of history, and the lost reality of the past. All themes overlap in some way, but none is a precise mirror image of the other. Many people confuse history with the unrecoverable past and confuse myth and memory with fiction.

Alex Roland captured the importance of the Apollo myth best: It is a retelling for a specific purpose. It is not as much history as it is "tribal rituals, meant to comfort the old and indoctrinate the young." He noted:

All the exhilarating stories are here: the brave, visionary young President who set America on a course to the moon and immortality; the 400,000 workers across the nation who built the Apollo spacecraft; the swashbuckling astronauts who exuded the right stuff; the preliminary flights of Mercury and Gemini—from Alan Shepard's suborbital arc into space, through John Glenn's first tentative orbits, through the rendezvous and spacewalks of Gemini that rehearsed the techniques necessary for Apollo. There is the 1967 fire that killed three astronauts and charred ineradicably the Apollo record and the Apollo memory; the circumlunar flight of Christmas 1968 that introduced the world to Earth-rise over the lunar landscape; the climax of Apollo 11 and Neil Armstrong's heroic piloting and modest words, "that's one small step for a man, one giant leap for mankind"; the even greater drama of Apollo 13, rocked by an explosion on the way to the moon and converted to a lifeboat that returned its crew safely to Earth thanks to the true heroics of the engineers in Houston; and, finally, the anticlimax of the last Apollo missions.

Roland finds an epic aura of Apollo in this recitation of the voyages of discovery. The missions, however, turned into a dead end rather than a new beginning, and no amount of heroic prose could overcome that plot twist.

Central to any discussion of Apollo, mythic event or not, requires the consideration of the role of prestige in originating and sustaining the effort. Prestige, for all of its ubiquity in the literature of human spaceflight, is

an imprecise term, and it perhaps obscures more than it illuminates. At sum, it signifies a demonstration of American superiority. But this superiority has many facets and audiences. It elicits both a "gut-level" reaction and calls for a more sophisticated explication. It is driven by politics of many sorts—international, bureaucratic, and domestic—none of them sufficient on their own to explain the primacy of human spaceflight in American culture, but all complexly intertwined.

Pride at home and prestige abroad were overwhelmingly significant rationales for undertaking an expansive Moon-landing program in the 1960s. In addition, there may well be four distinct attributes of the pride and prestige issue in Apollo. These include prestige on the international stage, using Apollo as a means for enhancing the attitudes of non-Americans toward the United States; pride at the national level, drawing the nation and its many peoples, priorities, and perspectives together; defining national identity, offering important ingredients into the national narrative celebrating exceptionalism among all else in the world; and embracing the idea of progress, and using the Apollo program as a symbol for American forward-thinking.

Almost from the beginning of thought about the potential of flight in space, theorists believed that the activity would garner worldwide prestige for those accomplishing it. For example, in 1946 the newly established RAND Corporation published the study *Preliminary Design of an Experimental World-Circling Spaceship*. This publication explored the viability of orbital satellites and outlined the technologies necessary for its success. Among its many observations, its comment on the prestige factor proved especially prescient: "A satellite vehicle with appropriate instrumentation can be expected to be one of the most potent scientific tools of the 20th Century. The achievement of a satellite craft would produce repercussions comparable to the explosion of the atomic bomb." In a paper published nine months later, RAND's James Lipp expanded on this idea: "Since mastery of the elements is a reliable index of material progress, the nation which first makes significant achievements in space travel will be acknowledged as the world leader in both military and scientific techniques. To visualize the impact on the world, one can imagine the consternation and admiration that would be felt here if the United States were to discover suddenly that some other nation had already put up a successful satellite."

This perspective is a classic application of what analysts have often referred to as "soft power." Coined by Harvard University professor Joseph Nye, the term gave a name to an alternative to threats and other forms of hard power in international relations. As Nye contends:

Soft power is the ability to get what you want by attracting and persuading others to adopt your goals. It differs from hard power, the ability to use the carrots and sticks of economic and military might to make others follow your will. Both hard and soft power are important . . . but attraction is much cheaper than coercion, and an asset that needs to be nourished.

In essence, activities such as Apollo represented a form of soft power, the ability to influence other nations through intangibles such as an impressive show of technological capability. It granted to the nation achieving it first, rightly as James Lipp forecast, an authenticity and gravitas not previously enjoyed among the world community. At root, this was an argument buttressing the role of spaceflight as a means of enhancing prestige on the world stage.

Even so, few appreciated the potential of spaceflight to enhance national prestige until the Sputnik crisis of 1957–58. Some have characterized that event as akin to Pearl Harbor, creating an illusion of a technological gap and providing the impetus for increased spending for aerospace endeavors, technical and scientific educational programs, and the chartering of new federal agencies to manage air and space research and development. This Cold War rivalry with the Soviet Union provided the key that opened the door to aggressive space exploration, not as an end in itself but as a means to achieving technological superiority in the eyes of the world. From the perspective of the 21st century it is difficult to appreciate the importance of the prestige factor in national thinking at the time.

John F. Kennedy responded to a perceived challenge of the Soviet Union by announcing the Apollo decision in 1961, and that rivalry sustained the effort. Kennedy put the world on notice that the United States would not take a back seat to its superpower rival. Kennedy said in 1962 that "we mean to be a part of it [spaceflight]—we mean to lead it. For the eyes of the world now look into space, to the moon and to the planets

beyond, and we have vowed that we shall not see it governed by a hostile flag of conquest, but by a banner of freedom and peace. We have vowed that we shall not see space filled with weapons of mass destruction, but with instruments of knowledge and understanding." Apollo was a contest of wills, of political systems, of superpowers. And the United States intended to win it. Lyndon Johnson summed this up well with his assertion, "Failure to master space means being second best in every aspect, in the crucial area of our Cold War world. In the eyes of the world first in space means first, period; second in space is second in everything."

There is no question but that the Apollo program in particular, but also all of the human spaceflight efforts of the United States, was firstly about establishing US primacy in technology and with it the pride and prestige of great success. Apollo served as a surrogate for war, challenging the Soviet Union head on in a demonstration of technological virtuosity. The desire to win international support for the "American way" became the *raison d'être* for the Apollo program, and it served that purpose far better than anyone imagined when first envisioned. Apollo became first and foremost a Cold War initiative and aided in demonstrating the mastery of the United States before the world. This may be seen in a succession of Gallup polls conducted during the 1960s in which the question was asked: "Is the Soviet Union ahead of the U.S. in space?" Until the middle part of the decade, about the time that the Gemini program began to demonstrate American prowess in space, the answer was always that the United States trailed the Soviets. By the height of the Apollo Moon landings, world opinion had shifted overwhelmingly in favor of the United States. The importance of Apollo as an instrument of US foreign policy—which is not necessarily identical with national prestige and geopolitics but is closely allied—should not be mislaid in this discussion. It served, and continues to do so, as an instrument for projecting the image of a positive, open, dynamic American society abroad.

So for decades the United States launched humans into space for prestige, measured against similar Soviet accomplishments, rather than for practical scientific or research goals. This was in essence positive symbolism—each new space achievement acquired political capital for the United States, primarily on the international stage. As military analyst Caspar Weinberger noted in 1971, space achievements gave "the people of the world an equally needed look at American superiority."

At the same time Apollo heightened national pride. Observers have long recognized the impact that science and technology plays in building confidence in the US system in the international arena. Neither scholars nor politicians, however, have explicitly appreciated the importance of science and technology in building public confidence at home. Science and technology has been used to build public confidence in government in the same way that presidents use foreign policy to salvage their approval ratings. It can be argued that science and technology offer even greater opportunities for building public confidence than foreign adventures. It does so because scientific and technological programs tend to be driven by a clearly defined mission and a large degree of program autonomy that helps to close off much interest-group politics, factors that tend to favor program success.

Apollo, and perhaps spaceflight more generally, conjured images of the best in the human spirit and served, in the words of journalist Gregg Easterbrook, as "a metaphor of national inspiration: majestic, technologically advanced, produced at dear cost and entrusted with precious cargo, rising above the constraints of the earth." National pride served almost as much of a motivator of the US commitment to Apollo as the quest for global power. It may well be that space achievements, particularly those involving direct human presence, remain a potent source of national pride and that such pride is the underpinning reason why the American public continues to support human spaceflight and would find a decision to end the spaceflight program unacceptable. Certainly, space images—an astronaut on the Moon—rank just below the American flag and the bald eagle as patriotic symbols. The self-image of the United States as a successful nation is threatened when we fail in our space efforts, as we have seen from the embarrassment of a misshaped mirror on the Hubble Space Telescope to the sense of collective loss when some of our best citizens die before our eyes in space-shuttle accidents. Americans expect a successful program of human spaceflight as part of what the United States does as a nation. They are not overly concerned with the content or objectives of specific programs. But they are concerned that what is done seems worth doing and is done well. That sense of pride in space accomplishment has been missing in recent years.

It is somewhat trite to suggest that America was founded on the idea of progress and that it remains both an amorphous concept and one central

to American national identity. In the 1830s an astute French interpreter of American society, Alexis de Tocqueville, observed that Americans had a "lively faith in human perfectibility." The manner in which these ideas have evolved over time has changed in relation to the larger society, and space exploration, especially Apollo, evinced these cherished conceptions.

Apollo had shown that virtually anything was possible. It suggested that America had both the capability and the wherewithal to accomplish truly astounding goals. All it needed was the will. As Senator Abraham Ribicoff mused in 1969, "If men can visit the Moon—and now we know they can—then there is no limit to what else we can do. Perhaps that is the real meaning of Apollo 11."

Space advocates also foresaw a new era of peace and mutual understanding arising in response to Apollo. Carl Sagan described "the unexpected final gift of Apollo" as "the inescapable recognition of the unity and fragility of the Earth." He added, "I'm struck again by the irony that spaceflight—conceived in the cauldron of nationalist rivalries and hatreds—brings with it a stunning transnational vision. You spend even a little time contemplating the Earth from orbit and the most deeply ingrained nationalisms begin to erode. They seem the squabbles of mites on a plum." Peaceful expansion beyond Earth, they thought, would lead to a golden age of peace and prosperity on Earth as national rivalries would be replaced with collective action toward a great positive goal. Forward-looking and aimed toward human perfections, Apollo signaled great opportunities for the future.

At its heart, this idea rested on what Richard T. Hughes has called the myth of the United States as a millennial nation, forever progressing to perfection. This sense of creating a perfect society has been present in American society from the very first. It also suggests that it is incumbent on those a part of this nation to further justice, equality, and liberty both inside and outside the confines of the United States. In many instances this is a positive set of attributes, as Hughes notes, but it might also be used to justify efforts "to export and impose its cultural and economics values throughout the world, regardless of the impact those policies might have on poor and dispossessed people in other parts of the world."

This sense of progress also embraced another national myth, that of the innocent nation. Without firm justification, the United States has come to believe that whatever it does is just and righteous and

representative of best that humanity has to offer the world. This may be seen in virtually all periods of American history, but it is especially present in the great struggles of the 20th century. World Wars I and II especially led Americans to believe they were fighting for the survival of all that was good against forces of evil. But it also may be seen in the Cold War against the Soviet Union and in the aftermath of 9/11 in the global war on terrorism. So too, Apollo for most Americans is a representation of progress and eventual perfection.

There is a fine line between national pride and jingoism. Through the decade of the 1960s, national pride dominated much of the discussion of Apollo, even penetrating to popular culture. The late social commentator and comedian Sam Kinison once said to other nations seeking to undertake space spectacles: "You really want to impress us! Bring back our Flag!" This is a particularly explicit expression of jingoism in relation to Apollo. While Kinison's challenge symbolized for many citizens US superiority and at the same time signaled the inferiority of all others, NASA situated its Apollo aspirations within the arena of international inclusiveness. If anything, these contradictory themes of national uniqueness and international cooperation have only intensified over time. This has long been critical to understanding the development of the space age.

Even without the crass jingoism of comedians, Apollo expresses those ideas quite effectively. Allusions to the "eagle" landing on the Moon, the placement of the US flag (and no other) on its surface, and the particular sense of possession Americans feel for the endeavor may be comfortable in the context of domestic affairs but are certainly increasingly uncomfortable as the space race recedes into history. Even so, the belief that humans would no longer be trapped on one tiny world, helpless to escape whatever might befall it, proved a powerful draw for those embracing spaceflight. As such, Apollo is celebrated as the first step off this world. Deeply mythic and powerfully nostalgic, Apollo's recollection in this manner resonates with the mythic frontier of the past that now morphs into the future frontier of space, a uniquely American perspective on this recent episode in history.

In part to advance the dominant narrative of Apollo as an American triumph, in part for mundane political reasons (such as expenditures in states where elections were up for grabs), in part to capture some of the aura of the Moon landings for a successive administration, and in part for

a sense of the need to invest in science and technology because of the very real benefits that accrue through the process, several presidents and their political allies have advocated for a return to the Moon. These decisions deserve mention here as indications of the ongoing hold Apollo and the Moon has on the public policy of the United States.

Even before the end of the Apollo program in the early 1970s, some Americans were encouraging a return to the Moon to establish a lunar base. When Richard Nixon took office, he appointed the Space Task Group to study post-Apollo plans and make recommendations. Chartered on February 13, 1969, under the chairmanship of Vice President Spiro T. Agnew, this group met throughout the spring and summer to plot a course for the post-Apollo space program. The politics of this effort was intense. NASA lobbied hard with the group, and especially its chair, for a far-reaching post-Apollo space program that included development of a space station, a reusable Space Shuttle, a Moon base, and a human expedition to Mars. The NASA position was well reflected in the group's report on September 15, 1969, but Nixon did not act on the Space Task Group's recommendations. Instead, he was silent on the future of the US space program for several months and in the end did not accept the recommendation. Nixon issued a March 7, 1970, statement that clearly announced his approach toward dealing with NASA and space exploration: "We must also recognize that many critical problems here on this planet make high priority demands on our attention and our resources." Instead of continued lunar exploration, he approved a Space Shuttle only for Earth orbital operations.

This episode of post-Apollo planning demonstrates a negative impact of the Moon landings—an object lesson on how not to develop and sustain a long-term strategic program of human space exploration. It seems that in part because of the pressures to achieve Apollo on the schedule mandated by JFK, the technical solutions were not sustainable. At some level the program may also not have been sustainable because it was so exceptional. Even though pro-space advocates understood this problem, the political priority of Apollo thwarted the implementation of future expansive exploration efforts.

One might conceivably draw several conclusions from this episode. First, those leading NASA have a vision of what should be undertaken in human spaceflight that is always more expansive and less popular

than what society as a whole, represented by the political leadership of the United States, is willing to undertake. Second, the technological challenge of bringing a new human space vehicle to reality is always greater than perceived at the time of the program's approval. Third, overpromising what might result from a particular program always has significant negative repercussions. That was certainly the story of any effort to return to the Moon, and we have seen it in numerous other settings as well.

The central question for NASA at the successful completion of Project Apollo was how best to continue its overall space exploration mission. This was a consideration especially important at the time because of the environment in Washington. As Apollo reached fruition, the nation was consumed with other crises: urban unrest, race riots, the Vietnam conflict and the antiwar movement, political radicalism of the left and the right, economic recession, welfare problems, and runaway budgets. Most important, the nation began to reassess the complex debate over the consensus of a vision of America as it had been articulated since the 1930s. This sustained criticism of national character and meaning plunged the United States into a fundamental divide over how to view the past, including Apollo. Political turmoil and activism led to the advance of a conservative perspective on politics, history, economics, and society, which emerged full-blown in the presidency of Ronald Reagan in the 1980s.

As this happened, Republicans in the White House—who served 26 of the years since the end of the Apollo program in 1972—embraced the legacy of Apollo and sought to replicate it on more than one occasion. Twenty years to the day after the Apollo 11 Moon landing, for example, Reagan's successor, President George H. W. Bush, made a Kennedy-like announcement by unveiling the ambitious Space Exploration Initiative (SEI) intended to return Americans to the Moon, establish a lunar base, and, then, using the space station and the Moon, reach Mars by the early part of the 21st century. The price tag for this effort was estimated at a whopping \$400 billion over two decades, although some estimated the costs as high as \$1 trillion. When briefed on this initiative, Stephen Kohashi, an aide to Senator Jake Garn (R-UT), exclaimed, "Have you lost your mind?" This demonstrated the constant push and pull between the desire of making measured, incremental progress in space exploration and aggressive funding increases that would both feed the NASA infrastructure and allow for aggressive Apollo-like programs.

Congress immediately reacted negatively. In votes for FY 1991 NASA funding, the SEI proposal was virtually zeroed out despite lobbying from Vice President Dan Quayle as the head of the National Aeronautics and Space Council, an advisory group to the president. Although the president castigated Congress for not "investing in America's future," members believed such a huge sum could be better spent elsewhere. Normally a strong supporter of NASA efforts, Maryland Senator Barbara Mikulski bluntly declared, "We're essentially not doing Moon-Mars."

In his support of SEI, Bush attempted to bring the space program full circle back to the early 1960s, but without the complementary elements that had made Kennedy's Apollo decision viable—that is, a crisis atmosphere that fostered the political will to do something spectacular, a favorable economic and technological climate, and tangible public support. In February 1993, looking for ways to cut the federal budget and thereby ease the federal deficit, the new Clinton administration announced that SEI did not fit into its plans for the space program. Accordingly, NASA Administrator Daniel S. Goldin said that the agency was not giving up on the idea of sending Americans back to the Moon and on to Mars, but simply "putting it off until we're ready and the nation is able to afford it." The fiscal year 1994 budget, therefore, contained no funding for SEI.

On January 14, 2004, President George W. Bush performed essentially a reenactment of his father by announcing a "Vision for Space Exploration" that called for humans to reach for the Moon and Mars during the next 30 years. As stated at the time, the fundamental goal of this vision was to advance US scientific, security, and economic interests through a robust space exploration program. In so doing, the president called for completion of an ongoing International Space Station (ISS) and retirement of the Space Shuttle fleet by 2010. Resources expended there would then go toward creating the enabling technologies necessary to return to the Moon and eventually to Mars. He also proposed a small increase in the NASA budget to help make this a reality. By 2008, however, it had become highly uncertain that the initiative could be realized, and in 2010 the program was cancelled by Bush's Democratic successor in the White House, Barack Obama. More recently, in December 2017 President Donald Trump directed NASA to "lead an innovative space exploration program to send American astronauts back to the moon, and

eventually Mars." Trump added: "The directive I am signing today will refocus America's space program on human exploration and discovery. It marks an important step in returning American astronauts to the moon for the first time since 1972 for long-term exploration and use." He commented that this time a Moon program would lead to a permanent presence rather than just flags and footprints left behind. So far, however, there has been but a modest investment in making this objective a reality.

When President Trump made these statements, he was playing off a widely shared continuing desire to go to the Moon. The inertia and especially the cost soon lead to the realization, though, that this desire is not sufficiently strong to foster much in the way of action. Perhaps we have seen replays of the criticism that arose almost immediately with the Apollo decision in 1961. At that time, critics both on the political left and right questioned the huge investment that went into making the Moon landings a reality, despite whatever magnificence it might represent for the nation as a whole. With a price tag of \$25.4 billion over an 11-year period, and more than \$180 billion in 2017 dollars, it was the most expensive undertaking of the United States government in the 1960s except for national security. Opponents on both the left and right have asserted since the Kennedy administration that Apollo was inherently wasteful and should not be repeated. For example, William Proxmire attacked NASA's efforts to extend Apollo by undertaking space colonization, commenting, "It's the best argument yet for chopping NASA's funding to the bone. . . . I say not a penny for this nutty fantasy." Many conservatives likewise questioned the role of government in engaging in such exploits and believed that the money might be spent more productively elsewhere or given back to taxpayers as a reduction.

The euphoria over successfully completing the Moon landings has established such a powerful memory that most people in the United States reflecting on it believe that NASA enjoyed enthusiastic support during the 1960s and that somehow the agency lost its compass thereafter. Contrarily, at only one point before the Apollo 11 mission, October 1965, did more than half of the public favor the lunar landing program. Americans of the era consistently ranked spaceflight near the top of those programs to be cut in the federal budget. Such a position is reflected in public opinion polls taken in the 1960s when the majority of Americans

ranked the space program as the government initiative most deserving of reduction, and its funding redistributed to Social Security, Medicare, and numerous other programs. While most Americans did not oppose space exploration per se, they certainly questioned spending on it when other problems appeared more pressing.

Since the heyday of Apollo, little has changed in this support for NASA and its human space exploration agenda. Many on the left view spaceflight, usually characterized as the human space program only, as a waste of resources that might be more effectively deployed to support other good ends. Many find themselves nodding in agreement when Josh Lyman, the White House assistant chief of staff in the fictional *West Wing* television series, told NASA officials in 2004 that his one priority for the space agency was that it stay out of the newspapers with tales of mismanagement and woe. He added that his agenda included using precious federal funds here on Earth to help people rather than to "conquer space."

Criticism of efforts to return to the Moon has taken myriad turns within the space community itself, as losers in the debate question the course taken. A subtext of this is that political decisions, especially Richard Nixon's decision to forego an expansive post-Apollo effort leading to a Moon base in the 1970s, lead directly to a conversation over the place of Apollo and a possible return to the Moon in the 21st century. As political scientist Howard E. McCurdy remarked, the Moon landing "was to America what the pyramids were to Egypt. It's one of our great accomplishments. . . . But when you go back and look, there were people, at the time who are expressing public misgivings. And in private—where you can get those kinds of conversations—[they] are pulling their hair out about this program."

The Moon landings were the result of a unique age in which the circumstances of politics, Cold War rivalries, the maturation of science and technology, a public willingness to embrace the endeavor, and the happenstance of an unfortunate presidential assassination converged to make possible the successful Apollo Moon landings. It seems unlikely that America will see another set of circumstances converge to create an environment conducive to a decision to send Americans to the Moon as a major national effort. There might be other ways in which humans might return there: perhaps a space race among other nations or

perhaps a commercial rationale for a human expedition to return to the Moon, but otherwise it seems unlikely as we look toward the future of lunar exploration in the first part of the 21st century.

It is impossible to be unimpressed by the efforts made to reach the Moon in the 1960s and to revel in the success of humankind in achieving this striking accomplishment. A spring 1999 poll of opinion leaders sponsored by leading news organizations in the United States, for example, ranked the 100 most significant news events of the 20th century. The Moon landings came in a very close second to the splitting of the atom and its use during World War II. Some found the process of deciding between these various events difficult. "It was agonizing," CNN anchor and senior correspondent Judy Woodruff said of the selection process. The eminent historian and public intellectual Arthur M. Schlesinger Jr. summarized the position of many opinion leaders. "The one thing for which this century will be remembered 500 years from now was: This was the century when we began the exploration of space," he commented. Schlesinger said he looked forward toward a positive future and that prompted him to rank the lunar landing first. "I put DNA and penicillin and the computer and the microchip in the first 10 because they've transformed civilization. Wars vanish," he said. "Pearl Harbor will be as remote as the War of the Roses," he said, referring to the English civil war of the 15th century. He added, "The order is essentially very artificial and fictitious. It's very hard to decide the atomic bomb is more important than getting on the Moon."

In such a situation, Apollo increasingly seems to be something Americans did once upon a time for reasons that have receded far into the background. Returning to the Moon certainly will not happen again anytime soon without a major realignment of rationales, needs, and priorities. In 100 years, Apollo may be remembered as a singular event, glorious and revered but viewed increasingly as an undertaking without lasting significance. While advocates of human space exploration view the astronauts who landed on the Moon as people akin to 15th-century seafarers such as Christopher Columbus, the vanguards of sustained human exploration and migration, this opinion seems to diminish with every passing year. Early visions of human exploration were entertaining, but they were rooted in a relative lack of understanding about the nature

of the Moon and planets and a view of colonization outmoded even at that time. The tradition of human space exploration may well continue, but probably not in the ways envisioned at the time of Apollo.

Some have characterized the Americans who fought and won World War II as "the greatest generation." Participants have been celebrated as heroes whether they carried a rifle in the Normandy invasion or built B-17 bombers in a factory. At the same time, no one would think that World War II was a great positive time. The death and destruction are something not to be repeated if possible. It would not surprise me to see recollections of Apollo echo that experience; participants will be venerated by those who were not present for the endeavor but who recognize it as a great achievement that will not be repeated. The trajectory may change, but probably not until there is a return to the Moon by humans and a discovery of something there of great worth.